THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(d)** hypotonic to the cell	Water entered.
7	**(d)** accumulation of food	Phloem blocked.
8	**(c)** Fungi	Hyphae and chitin.
9	**(d)** 14 March	Mid cycle.
10	**(b)** 4000 J	400 W x 10 s.
11	**(c)** synthesis of ATP	Mitochondria do ATP.
12	**(d)** glandular epithelium	Secretion.
13	**(a)** Both A and R are true and R is the correct explanation of A.	ATP is the energy currency of the cell, which is why mitochondria are called powerhouses.
14	**(a)** Both A and R are true and R is the correct explanation of A.	Growth requires new cells, which are produced by dividing meristematic cells.
15	**(a)** Both A and R are true and R is the correct explanation of A.	Interbreeding with fertile offspring is the defining feature of a species.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	SA/V = 6 and 2 (1); Larger cell exchanges less efficiently, so it divides (1).	1+1
17	Secondary xylem accumulates every year (1); Lignified walls give strength (1).	1+1
18	New evidence from molecular studies (1); Discovery of new organisms (1).	1+1
19	Muscle 46, sperm 23 (1); Egg 23, zygote 46 (1).	1+1
20	Absorbed = 0.7 x 342 (1); = 239 W/m squared (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	0.5 per cent: gains mass (1); 5 per cent: loses mass (1); 1 per cent is isotonic with the potato cell sap (1).	1+1+1
22	Potato turns blue-black as it stores starch (1); Carrot gives less blue-black (1); Parenchyma stores food (1).	1+1+1
23	Feathers? yes: pigeon (1); No: has hair? yes: bat (1); No: gills with scales: shark; moist skin: frog; dry scaly skin: snake (1).	1+1+1
24	Ovulation about 14 days before the next period: day 16 (1); Fertile period: days 13 to 17 (1); Sperms survive 3 days before (1).	1+1+1
	OR: Annual emission = 250 x 12 x 0.8 = 2400 kg (1); Trees = 2400/22 (1); = 109 trees (1).	1+1+1
25	Break cells gently (1); Centrifuge at low speed to remove nuclei (1); Higher speed to get mitochondria (1).	1+1+1
	OR: Constant energy demand (1); Continuous contraction (1); Supplies ATP (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) Osmosis is the movement of water molecules through a selectively permeable membrane from a region of higher water concentration to a region of lower water concentration (1). (ii) Water enters the raisin cells by osmosis because the surrounding water is hypotonic to the cell contents (1). (iii) The strip in the salt solution (1). (iv) Salt solution: -10% (loses water); distilled water: $+12\%$ (gains water) (1); the cell loses water in a hypertonic medium and gains water in a hypotonic medium (1).	1+1+1+2
27	Cell type: prokaryote - bacterium (1); Eukaryote unicellular: Amoeba (no wall), yeast (wall) (1); Multicellular plants: vascular? no: moss; yes: seeds? no: fern (1); Seeds: cones - pine; flowers - rose (dicot), wheat (monocot) (1); Characters: cell wall, vascular tissue, seeds, cotyledon (1).	1+1+1+1+1
	OR: Tube 1: moist cotton, room temperature (1); Tube 2: dry cotton (1); Tube 3: seeds submerged in boiled water with oil (1); Tube 4: moist cotton in a fridge (1); Only tube 1 germinates; conclusion (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** cell wall	Rigid wall.
2	**(d)** sensory neuron, relay neuron and motor neuron	Reflex arc.
3	**(d)** animal	Animalia.
4	**(a)** wings or hairs	Light structures.
5	**(d)** the ground is heated by sunlight and warms the air	Surface heating.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(a)** lignin	Lignified walls.
8	**(a)** dolphins have lungs, hair and mammary glands	Mammalia vs Pisces.
9	**(a)** oviduct	Tube.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
11	**(a)** hypertonic to A and hypotonic to B	Different internal concentrations.
12	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
13	**(a)** Both A and R are true and R is the correct explanation of A.	With one parent and no gamete fusion, the offspring are clones of the parent.
14	**(a)** Both A and R are true and R is the correct explanation of A.	Albedo is the fraction of sunlight reflected; snow reflects a large fraction, so its albedo is high.
15	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the composition of the wall does not explain why animal cells lack it.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	It protects the cell and controls entry and exit of substances (1); It allows only some substances to pass (1).	1+1
17	Xylem upward only, phloem both directions (1); Xylem passive; phloem needs energy (1).	1+1
18	Homeothermic: birds and mammals (1); Poikilothermic: fish, amphibia, reptiles (1).	1+1
19	Pollen grains carry male gametes (1); Ovule has the egg; becomes seed after fertilisation (1).	1+1
20	Long coastline and monsoon dependence (1); Large population in agriculture (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Moss: gametophyte dominant (1); Fern: sporophyte dominant (1); Both need water for fertilisation (1).	1+1+1
22	Seeds from sexual reproduction give variation (1); Cutting is a clone (1); Mitosis vs meiosis and fertilisation (1).	1+1+1
23	Dry air has little water vapour to trap heat (1); Quick cooling at night (1); Bare sand heats up fast (1).	1+1+1
24	Fill the bag with starch and glucose (1); Place in water (1); Glucose passes out; starch stays (1).	1+1+1
	OR: Vascular cambium: secondary xylem and phloem (1); Cork cambium: cork (1); Together form secondary growth (1).	1+1+1
25	Include unrelated organisms (1); Based on a few characters (1); Different evolutionary origins (1).	1+1+1
	OR: Unisexual flowers (1); Anthers and stigma mature at different times (1); Different positions (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) Intercalary meristem (1). (ii) Lateral meristem (cambium) (1). (iii) Apical meristem (1). (iv) The intercalary meristem is at the base of the leaves or internodes, below the cut, so it is not removed (1), whereas the apical meristem at the tip is removed when the tip is cut (1).	1+1+1+2
27	Xerophytes: thick cuticle, sunken stomata (1); Hydrophytes: aerenchyma (1); Mesophytes: normal tissues (1); Root tissues for water (1); Support tissues (1).	1+1+1+1+1
	OR: Roots to absorb water (1); Vascular tissue for transport (1); Seeds protect the embryo (1); Flowers for pollination (1); Fruits for dispersal (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** bacterial	Prokaryote.
2	**(d)** xylem is intact	Water still moves.
3	**(a)** Gymnosperms	Cones with naked seeds.
4	**(d)** 13 to 15 March	Around ovulation.
5	**(a)** water filtration and flood control	Natural sponge.
6	**(c)** large central vacuole	Takes up space.
7	**(c)** a single nucleus and spindle shape	Unstriated.
8	**(d)** belong to the same genus but different species	Same genus Canis.
9	**(a)** cambia of stock and scion are in contact	Tissue contact.
10	**(c)** absorption of UV by ozone	Ozone heating.
11	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
12	**(b)** dendrite to axon	Direction.
13	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true; the nature of xylem does not explain the function of phloem.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true; R states the rule of writing but does not explain why the human name is *Homo sapiens*.
15	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true but describe different organisms and methods.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Protoplasm separates from the cell wall (1); Gap between wall and membrane (1).	1+1
17	No growth in length or girth (1); No new organs would develop (1).	1+1
18	Characteristic of Chordata (1); Used to identify the groups (1).	1+1
19	IVF (test-tube baby) (1); Artificial insemination (1).	1+1
20	UV has the energy to break O2 and O3 bonds (1); Visible light has less energy (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	30 units are reflected and 70 are absorbed (1); The Earth radiates back 70 units in equilibrium (1); Greenhouse gases delay the escape of heat (1).	1+1+1
22	Root tip: mitosis, growth (1); Testis: meiosis for gametes (1); Chromosome numbers (1).	1+1+1
23	Dicot stem (1); Vascular bundles in a ring (1); Open bundles with cambium (1).	1+1+1
24	Fungi (1); Heterotrophic by absorption (1); Eukaryotic with a cell wall (unicellular like yeast) (1).	1+1+1
	OR: Frog: thousands of eggs, no care (1); Hen: few eggs, some care (1); Human: one baby, extensive care (1).	1+1+1
25	Saving = 20 W x 100 = 2 kW (1); Energy = 2 x 8 x 250 = 4000 kWh (1); CO2 = 3200 kg (1).	1+1+1
	OR: They contain chlorophyll (1); They trap sunlight (1); They make food (glucose) by photosynthesis (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

26	(i) *Mangifera* (1). (ii) *tigris* (1). (iii) No; the species name alone is not enough, they belong to different genera (1). (iv) *Panthera tigris*: genus begins with a capital letter, species with a small letter, and the name is in italics or underlined separately (1); scientific names are the same worldwide and avoid confusion caused by many local names (1).	1+1+1+2
27	Prokaryotic: no true nucleus, no membrane-bound organelles (1); Eukaryotic: nucleus and organelles (1); Eukaryotic cells divide the work among organelles (1); Viruses have no cell structure (1); Viruses multiply only inside host cells (1).	1+1+1+1+1
	OR: Dicot: ring of open bundles (1); Monocot: scattered closed bundles (1); Cambium present in dicots (1); No secondary growth in monocots (1); Examples (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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